



B737 MAX

LINE-FIT

**FLIGHT DECK ENTRY
VIDEO SURVEILLANCE
(FDEVSS)**

INTRODUCTION



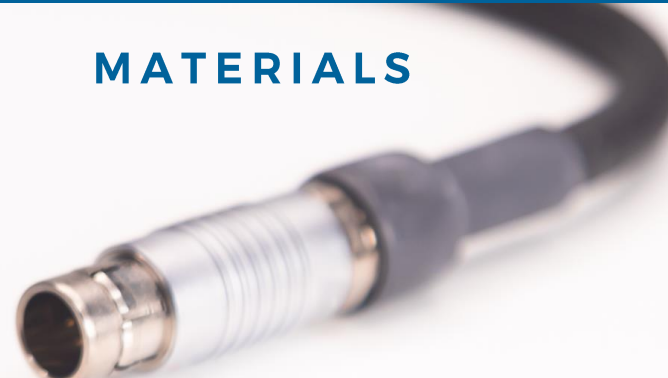
| ICAO



REGULATIONS



MATERIALS



INTRODUCTION



Satisfies the requirement for a non-intrusive, non-distracting method of identification by the Captain and First Officer of who is trying to gain access to the Flight Deck.

Acts as deterrent to terrorism and air rage and helps to prevent catastrophic loss of aircraft from terrorism or revenge motivation.

Represents low-cost option to significantly increase levels of aircraft safety and security.

System forms part of the Boeing 737 MAX Type Certificate and is certified wherever the B737 MAX is certified to fly.

This system has been developed in accordance with Boeing D6-83088 to interface with the Multi-Functional Displays which feature on Boeing aircraft.

There are over 700x AD Aerospace FDEVSS flying globally delivered direct to Boeing, major airlines, and lessors.

AD Aerospace is specified as part of the baseline configuration for most of the major lessors globally.

**OFFERABLE ON THE BOEING
737 MAX OPTIONS
CATALOGUE**



ICAO



REGULATIONS

Global regulators strongly recommend video systems...

Many European carriers proceeded with the installation of Flight Deck Entry Video Surveillance Systems after 9/11 and before mandates were issued.

ICAO Annex 6 – Part 1 – Chapter 13 – Standard 13.2.3(b) (1 Nov 2003):

"means shall be provided for monitoring from either pilot's station the entire door area outside the flight crew compartment to identify persons requesting entry and to detect suspicious behaviour or potential threat."

...interpreted into the Eurocae Minimum Operating Specification ED-123 which describes the minimum specification for a cockpit door surveillance system.



EU COMMISSION REGULATION (EC) No 859/2008 Sub-Part S OPS 1.1255 (20 Aug 2008):

"In all aeroplanes which are equipped with a flight crew compartment door...means shall be provided for monitoring from either pilot's station the area outside the flight crew compartment to the extent necessary to identify persons requesting entry to the flight crew compartment and to detect suspicious behaviour or potential threat."

MATERIALS

FV-0406 CAMERA



FV-0530 VIDEO SWITCH



FV-0950 CCU



FV-1070 CONTROL PANEL



MATERIALS

FV-0406 CAMERA

Video coverage for this type of area requires three cameras for the B737, one camera located directly above the flight deck door, one above the forward passenger door and the other above the galley door with their fields of view crossing.

The FV-0406 is a monochrome miniature video camera for general aerospace use.

The camera is designed to be fitted within the aircraft, in a pressurised, heated area. This may be above the ceiling panels, and may therefore be subject to water, and to greater temperature extremes than the main cabin.

The camera is designed to be panel mounted, and to view through the panel.

Functional down to 0.01 lux, which is significantly lower than the 1.0 lux cabin night-time aircraft cabin lighting SAE requirement, allowing the flight crew to see into a darkened cabin, and identify an individual with the cabin lights off.



Specifications	
Pick Up Element	¼ inch interline transfer CCD image sensor
Number of Pixels (EIA)	537 x 505
Scanning System	2:1 interlaced
Horizontal Resolution	380 TV lines per picture height
AGC	18dB
Electronic Iris	1/60 to 1/100,000 second shuttering
Video Output	1.0V p-p
Power consumption	4W (from CCU) (passive convection cooling)
Weight	50g
Size (camera head)	56mm square
Qualification	RTCA DO-160D
MTBF	100,000 hrs
MTBUR	80,000 hrs
Boeing Qualified	D6-36440 through TSA BTS-03-7905R2

MATERIALS

FV-0530 VIDEO SWITCH

AD Aerospace’s FV-0530 is an airborne switcher designed to control images from external and/or internal cameras installed on board an aircraft.

The unit has up to 4 balanced video inputs, which are switched to the various MFD screens in accordance with the Boeing D6-83088 “Multifunction Display Flight Deck Entry Video Surveillance System Requirements Document”.

The unit is used as a security feature, across all Boeing models, and is therefore designed with the restrictions of the SCSRD (Standard Cabin Systems Requirement Document) Boeing D6-36440 Rev C.

Unit power input is protected by a 1A anti-surge fuse. Unit will also provide 24V output for peripherals, protected by a 0.65A resettable fuse.



Specifications	
Video Inputs	4 composite video, 100 ohm balanced
Power Requirement	115V 400Hz AC at 50W maximum
Size	Mounting Plate 340 mm x 275 mm x100 mm
Weight	2.5kg
Qualification	RTCA DO-160D
MTBF	40,000 hrs
MTBUR	32,000 hrs
Boeing Qualified	D6-36440 through TSA BTS-05-8431

MATERIALS

FV-0950 CAMERA CONTROL UNIT



The Camera Interface Unit interfaces with the cameras via Camera Control Units.

The Camera Control Units provide regulated, filtered DC power to the cameras, as well as providing a 100-ohm balanced video signal drive output for distances of up to 100 meters (328 ft), ensuring a cleaner image.

One Camera Control Unit is required for each camera.

Specifications	
Video Inputs	1.0V p-p into 75 Ohms unbalanced
Video Output	2 off 1.0V p-p into 100 Ohm balanced line
Power Consumption	250mA @ 28V dc per RTCA DO-160D
Weight	240g
Qualification	RTCA DO-160D
MTBF	50,000 hrs
MTBUR	40,000 hrs
Boeing Qualified	D6-36440 through TSA BTS-03-7905R2

MATERIALS

FV-1070 CONTROL PANEL



The Video Control Panel is used in conjunction with the MFD Display Video Interface Unit.

The front panel switches are fitted behind a light plate which provides trans-illumination through the switches, around the panel and through the lettering, allowing the clear identification in a darkened flight deck.

Power for the light plate is provided from a 5V AC source external to the unit.

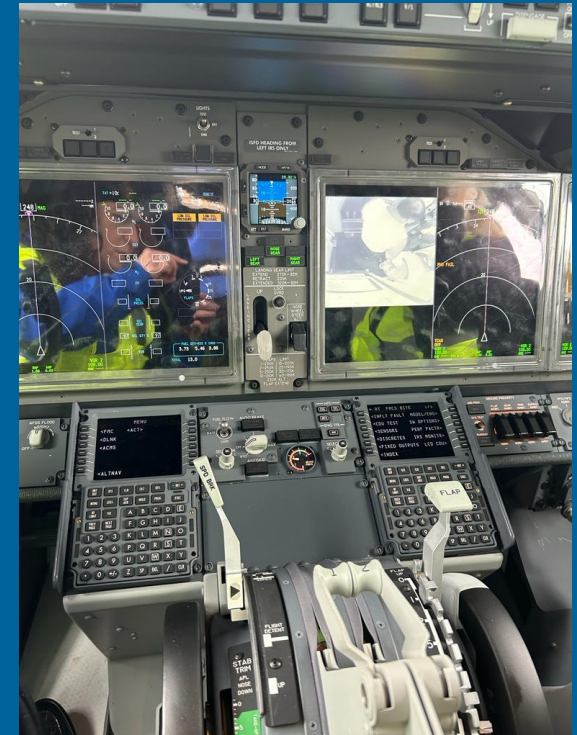
No power conditioning is provided on this input.

Specifications	
Size	146mm x 48mm
Weight	1kg
Materials	1.6mm Al Alloy Sheet
Finish	Alocrom to Def Stan 03-18 Paint & Knob colour per PIN number Background colour for Push-button switch is BAC706 Black
Mounting	4 off Dzus ¼ turn fasteners
Qualification	RTCA DO-160D
MTBF	100,000 hrs
MTBUR	80,000 hrs
Boeing Qualified	D6-36440 through TSA BTS-05-8431

B737 MAX FDEVSS CUSTOMERS



INSTALLATION



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